

Triple integrals in Spherical Coordinates

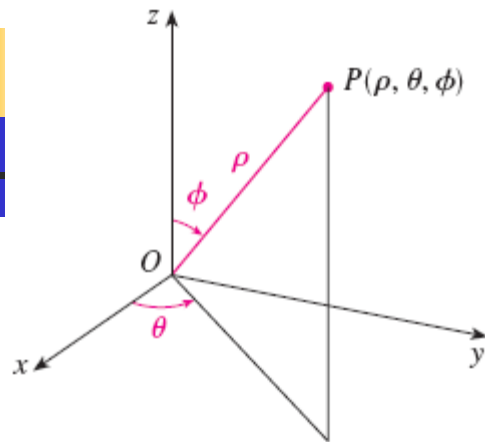


FIGURE 1

The spherical coordinates of a point

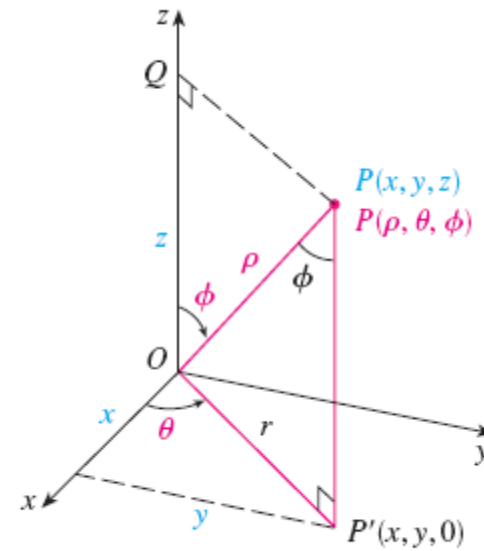


FIGURE 5

The relationship between rectangular and spherical coordinates can be seen from Figure 5. From triangles OPQ and OPP' we have

$$z = \rho \cos \phi \quad r = \rho \sin \phi$$

But $x = r \cos \theta$ and $y = r \sin \theta$, so to convert from spherical to rectangular coordinates, we use the equations

1

$$x = \rho \sin \phi \cos \theta \quad y = \rho \sin \phi \sin \theta \quad z = \rho \cos \phi$$

Also, the distance formula shows that

2

$$\rho^2 = x^2 + y^2 + z^2$$

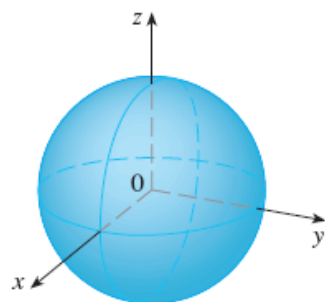


FIGURE 2 $\rho = c$, a sphere

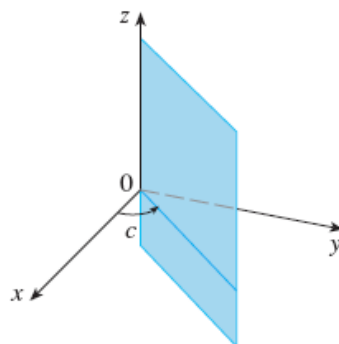
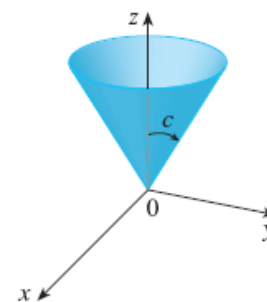
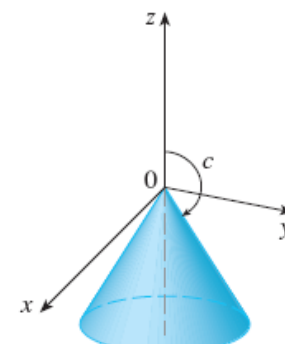


FIGURE 3 $\theta = c$, a half-plane



$$0 < c < \pi/2$$

FIGURE 4 $\phi = c$, a half-cone



$$\pi/2 < c < \pi$$

In the spherical coordinate system the counterpart of a rectangular box is a **spherical wedge**

$$E = \{(\rho, \theta, \phi) \mid a \leq \rho \leq b, \alpha \leq \theta \leq \beta, c \leq \phi \leq d\}$$

where $a \geq 0$ and $\beta - \alpha \leq 2\pi$, and $d - c \leq \pi$.

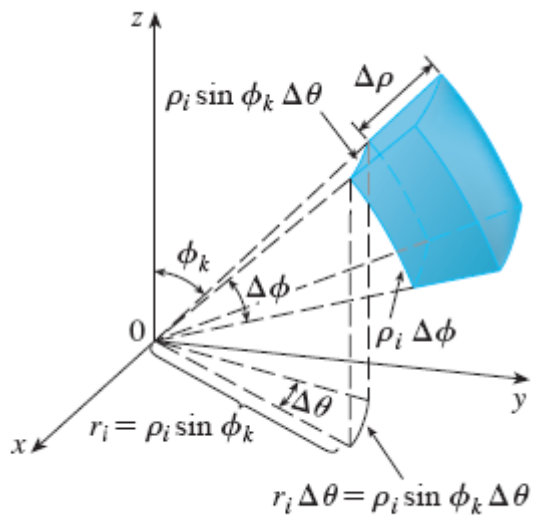


FIGURE 7

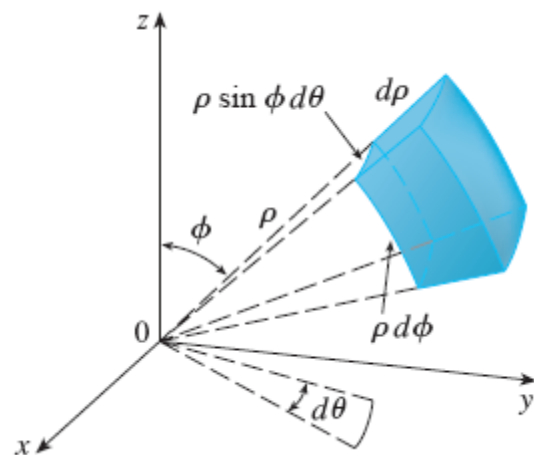


FIGURE 8

Volume element in spherical coordinates: $dV = \rho^2 \sin \phi \, d\rho \, d\theta \, d\phi$

$$\begin{aligned} \iiint_E f(x, y, z) \, dV &= \lim_{l, m, n \rightarrow \infty} \sum_{i=1}^l \sum_{j=1}^m \sum_{k=1}^n f(x_{ijk}^*, y_{ijk}^*, z_{ijk}^*) \Delta V_{ijk} \\ &= \lim_{l, m, n \rightarrow \infty} \sum_{i=1}^l \sum_{j=1}^m \sum_{k=1}^n f(\tilde{\rho}_i \sin \tilde{\phi}_k \cos \tilde{\theta}_j, \tilde{\rho}_i \sin \tilde{\phi}_k \sin \tilde{\theta}_j, \tilde{\rho}_i \cos \tilde{\phi}_k) \tilde{\rho}_i^2 \sin \tilde{\phi}_k \Delta \rho \Delta \theta \Delta \phi \end{aligned}$$

But this sum is a Riemann sum for the function

$$F(\rho, \theta, \phi) = f(\rho \sin \phi \cos \theta, \rho \sin \phi \sin \theta, \rho \cos \phi) \rho^2 \sin \phi$$

Consequently, we have arrived at the following **formula for triple integration in spherical coordinates**.

$$\begin{aligned} \text{3} \quad & \iiint_E f(x, y, z) dV \\ &= \int_c^d \int_\alpha^\beta \int_a^b f(\rho \sin \phi \cos \theta, \rho \sin \phi \sin \theta, \rho \cos \phi) \rho^2 \sin \phi d\rho d\theta d\phi \end{aligned}$$

where E is a spherical wedge given by

$$E = \{(\rho, \theta, \phi) \mid a \leq \rho \leq b, \alpha \leq \theta \leq \beta, c \leq \phi \leq d\}$$

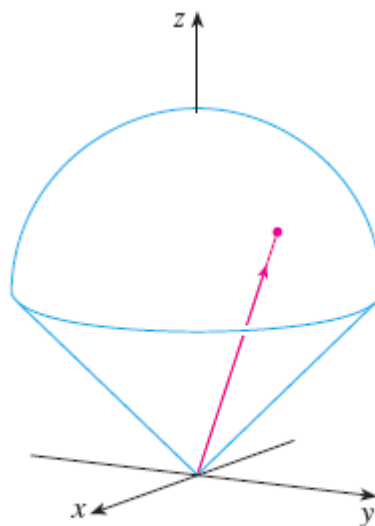
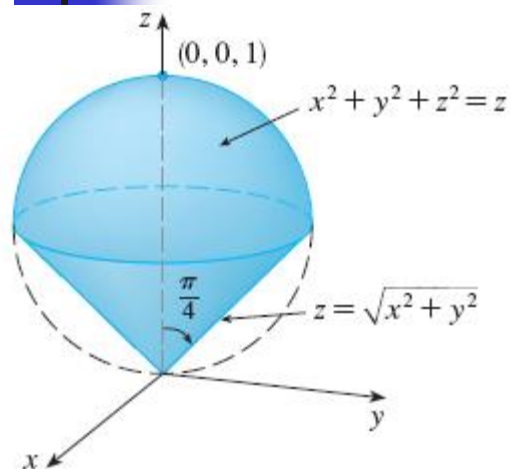
Example Evaluate $\iiint_B e^{(x^2+y^2+z^2)^{3/2}} dV$, where B is the unit ball:

$$B = \{(x, y, z) \mid x^2 + y^2 + z^2 \leq 1\}$$

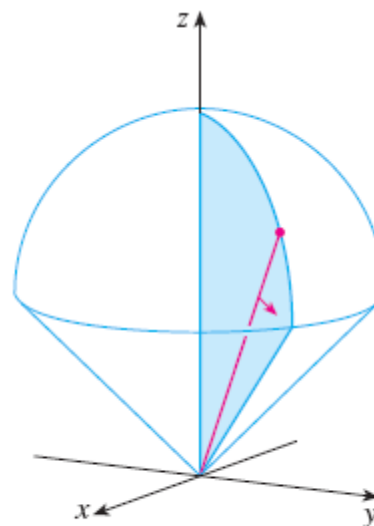
$$\frac{4}{3}\pi(e - 1)$$

V EXAMPLE 4 Use spherical coordinates to find the volume of the solid that lies above the cone $z = \sqrt{x^2 + y^2}$ and below the sphere $x^2 + y^2 + z^2 = z$. (See Figure 9.)

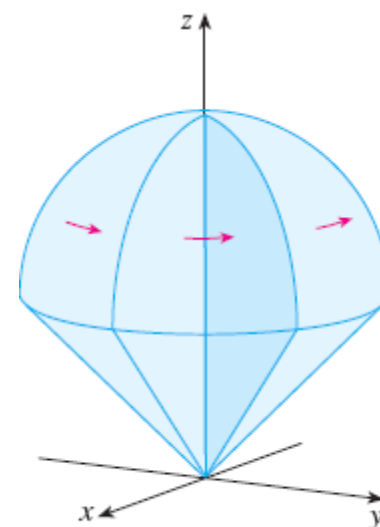
$$\frac{\pi}{8}$$



ρ varies from 0 to $\cos \phi$
while ϕ and θ are constant.



ϕ varies from 0 to $\pi/4$
while θ is constant.



θ varies from 0 to 2π .

Change of Variables in Multiple Integrals

In one-dimensional calculus we often use a change of variable (a substitution) to simplify an integral. By reversing the roles of x and u , we can write the Substitution Rule (4.5.5) as

1

$$\int_a^b f(x) \, dx = \int_c^d f(g(u)) g'(u) \, du$$

where $x = g(u)$ and $a = g(c)$, $b = g(d)$. Another way of writing Formula 1 is as follows:

2

$$\int_a^b f(x) \, dx = \int_c^d f(x(u)) \frac{dx}{du} \, du$$

More generally, we consider a change of variables that is given by a **transformation** T from the uv -plane to the xy -plane:

$$T(u, v) = (x, y)$$

where x and y are related to u and v by the equations

3

$$x = g(u, v) \quad y = h(u, v)$$

or, as we sometimes write,

$$x = x(u, v) \quad y = y(u, v)$$

We usually assume that T is a C^1 **transformation**, which means that g and h have continuous first-order partial derivatives.

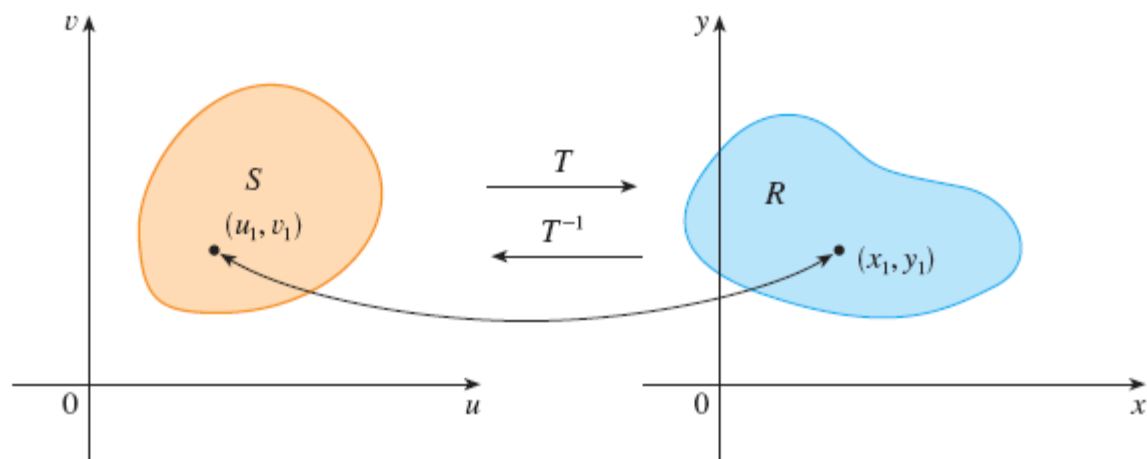
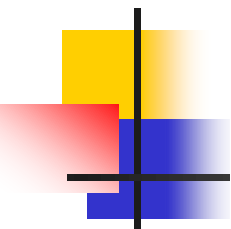


FIGURE 1

A transformation T is really just a function whose domain and range are both subsets of $\mathbb{R}^2$. If $T(u_1, v_1) = (x_1, y_1)$, then the point (x_1, y_1) is called the **image** of the point (u_1, v_1) . If no two points have the same image, T is called **one-to-one**. Figure 1 shows the effect of a transformation T on a region S in the uv -plane. T transforms S into a region R in the xy -plane called the **image of S** , consisting of the images of all points in S .

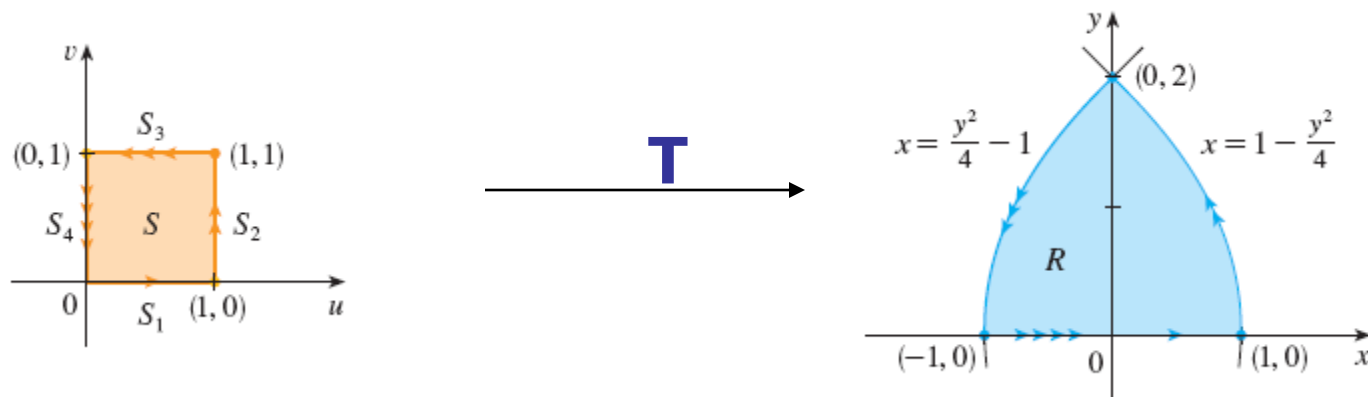
If T is a one-to-one transformation, then it has an **inverse transformation** T^{-1} from the xy -plane to the uv -plane and it may be possible to solve Equations 3 for u and v in terms of x and y :

$$u = G(x, y) \quad v = H(x, y)$$

V EXAMPLE 1 A transformation is defined by the equations

$$x = u^2 - v^2 \quad y = 2uv$$

Find the image of the square $S = \{(u, v) \mid 0 \leq u \leq 1, 0 \leq v \leq 1\}$.



Now let's see how a change of variables affects a double integral. We start with a small rectangle S in the uv -plane whose lower left corner is the point (u_0, v_0) and whose dimensions are Δu and Δv . (See Figure 3.)

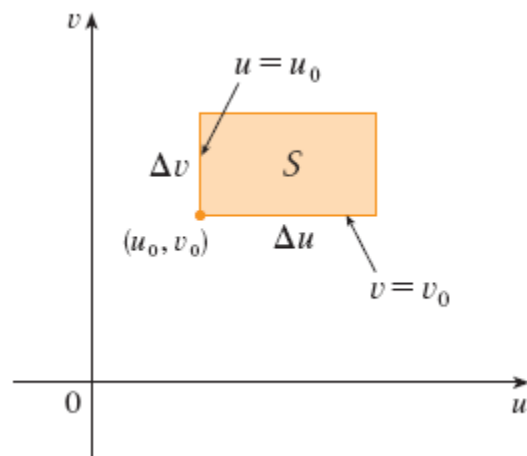


FIGURE 3

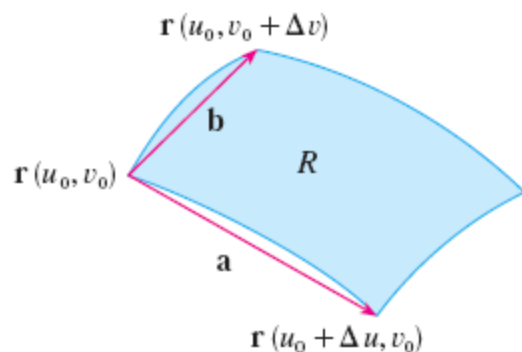
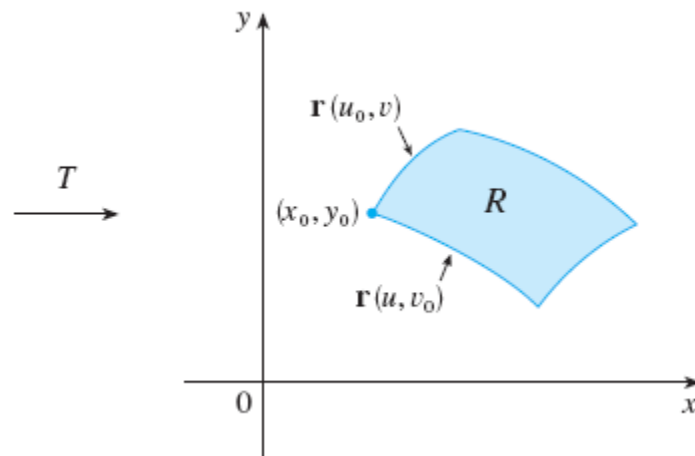


FIGURE 4

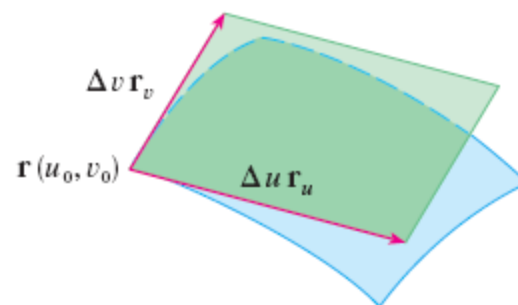


FIGURE 5

The image of S is a region R in the xy -plane, one of whose boundary points is $(x_0, y_0) = T(u_0, v_0)$. The vector

$$\mathbf{r}(u, v) = g(u, v) \mathbf{i} + h(u, v) \mathbf{j}$$

is the position vector of the image of the point (u, v) . The equation of the lower side of S is $v = v_0$, whose image curve is given by the vector function $\mathbf{r}(u, v_0)$. The tangent vector at (x_0, y_0) to this image curve is

$$\mathbf{r}_u = g_u(u_0, v_0) \mathbf{i} + h_u(u_0, v_0) \mathbf{j} = \frac{\partial x}{\partial u} \mathbf{i} + \frac{\partial y}{\partial u} \mathbf{j}$$

Similarly, the tangent vector at (x_0, y_0) to the image curve of the left side of S (namely, $u = u_0$) is

$$\mathbf{r}_v = g_v(u_0, v_0) \mathbf{i} + h_v(u_0, v_0) \mathbf{j} = \frac{\partial x}{\partial v} \mathbf{i} + \frac{\partial y}{\partial v} \mathbf{j}$$

We can approximate the image region $R = T(S)$ by a parallelogram determined by the secant vectors

$$\mathbf{a} = \mathbf{r}(u_0 + \Delta u, v_0) - \mathbf{r}(u_0, v_0) \quad \mathbf{b} = \mathbf{r}(u_0, v_0 + \Delta v) - \mathbf{r}(u_0, v_0)$$

shown in Figure 4. But

$$\mathbf{r}_u = \lim_{\Delta u \rightarrow 0} \frac{\mathbf{r}(u_0 + \Delta u, v_0) - \mathbf{r}(u_0, v_0)}{\Delta u}$$

and so

$$\mathbf{r}(u_0 + \Delta u, v_0) - \mathbf{r}(u_0, v_0) \approx \Delta u \mathbf{r}_u$$

Similarly

$$\mathbf{r}(u_0, v_0 + \Delta v) - \mathbf{r}(u_0, v_0) \approx \Delta v \mathbf{r}_v$$

This means that we can approximate R by a parallelogram determined by the vectors $\Delta u \mathbf{r}_u$ and $\Delta v \mathbf{r}_v$. (See Figure 5.) Therefore we can approximate the area of R by the area of this parallelogram, which, from Section 12.4, is

$$\boxed{6} \quad |(\Delta u \mathbf{r}_u) \times (\Delta v \mathbf{r}_v)| = |\mathbf{r}_u \times \mathbf{r}_v| \Delta u \Delta v$$

Computing the cross product, we obtain

$$\mathbf{r}_u \times \mathbf{r}_v = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \frac{\partial x}{\partial u} & \frac{\partial y}{\partial u} & 0 \\ \frac{\partial x}{\partial v} & \frac{\partial y}{\partial v} & 0 \end{vmatrix} = \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial y}{\partial u} \\ \frac{\partial x}{\partial v} & \frac{\partial y}{\partial v} \end{vmatrix} \mathbf{k} = \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{vmatrix} \mathbf{k}$$

The determinant that arises in this calculation is called the *Jacobian* of the transformation and is given a special notation.

7 Definition The **Jacobian** of the transformation T given by $x = g(u, v)$ and $y = h(u, v)$ is

$$\frac{\partial(x, y)}{\partial(u, v)} = \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{vmatrix} = \frac{\partial x}{\partial u} \frac{\partial y}{\partial v} - \frac{\partial x}{\partial v} \frac{\partial y}{\partial u}$$

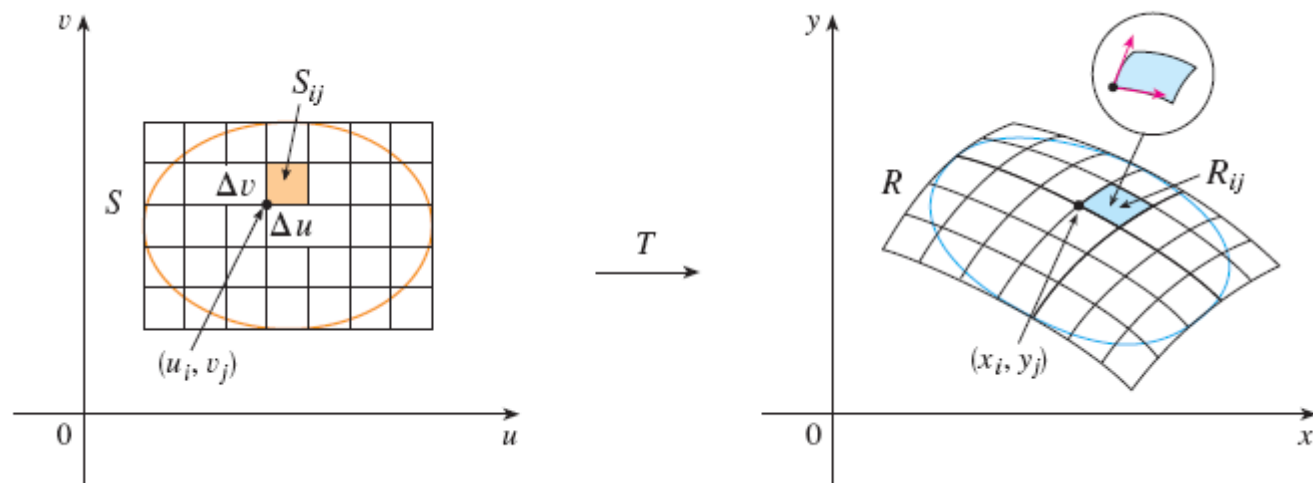


FIGURE 6

With this notation we can use Equation 6 to give an approximation to the area ΔA of R :

$$(8) \quad \Delta A \approx \left| \frac{\partial(x, y)}{\partial(u, v)} \right| \Delta u \Delta v$$

where the Jacobian is evaluated at (u_0, v_0) .

Next we divide a region S in the uv -plane into rectangles S_{ij} and call their images in the xy -plane R_{ij} . (See Figure 6.)

Applying the approximation (8) to each R_{ij} , we approximate the double integral of f over R as follows:

$$\begin{aligned} \iint_R f(x, y) dA &\approx \sum_{i=1}^m \sum_{j=1}^n f(x_i, y_j) \Delta A \\ &\approx \sum_{i=1}^m \sum_{j=1}^n f(g(u_i, v_j), h(u_i, v_j)) \left| \frac{\partial(x, y)}{\partial(u, v)} \right| \Delta u \Delta v \end{aligned}$$



where the Jacobian is evaluated at (u_i, v_j) . Notice that this double sum is a Riemann sum for the integral

$$\iint_S f(g(u, v), h(u, v)) \left| \frac{\partial(x, y)}{\partial(u, v)} \right| du dv$$

The foregoing argument suggests that the following theorem is true. (A full proof is given in books on advanced calculus.)

9 Change of Variables in a Double Integral Suppose that T is a C^1 transformation whose Jacobian is nonzero and that maps a region S in the uv -plane onto a region R in the xy -plane. Suppose that f is continuous on R and that R and S are type I or type II plane regions. Suppose also that T is one-to-one, except perhaps on the boundary of S . Then

$$\iint_R f(x, y) dA = \iint_S f(x(u, v), y(u, v)) \left| \frac{\partial(x, y)}{\partial(u, v)} \right| du dv$$

An Example

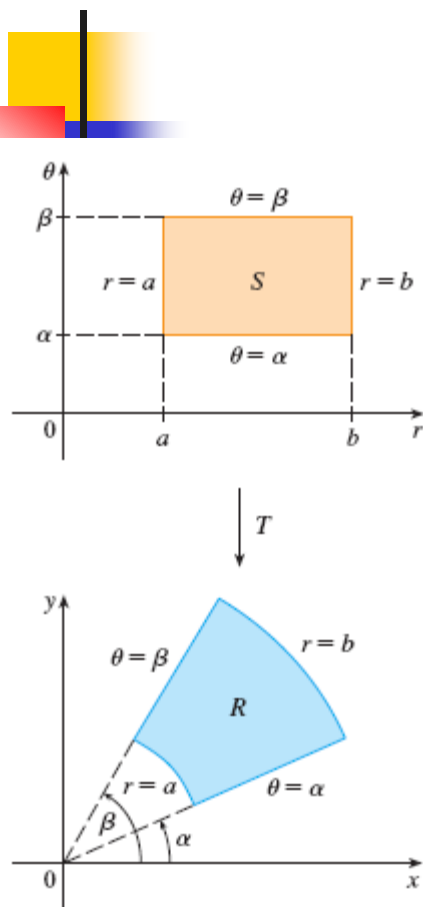


FIGURE 7

The polar coordinate transformation

$$x = g(r, \theta) = r \cos \theta \quad y = h(r, \theta) = r \sin \theta$$

and the geometry of the transformation is shown in Figure 7. T maps an ordinary rectangle in the $r\theta$ -plane to a polar rectangle in the xy -plane. The Jacobian of T is

$$\frac{\partial(x, y)}{\partial(r, \theta)} = \begin{vmatrix} \frac{\partial x}{\partial r} & \frac{\partial x}{\partial \theta} \\ \frac{\partial y}{\partial r} & \frac{\partial y}{\partial \theta} \end{vmatrix} = \begin{vmatrix} \cos \theta & -r \sin \theta \\ \sin \theta & r \cos \theta \end{vmatrix} = r \cos^2 \theta + r \sin^2 \theta = r > 0$$

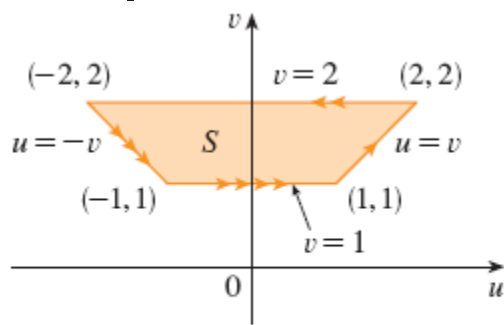
Thus Theorem 9 gives

$$\begin{aligned} \iint_R f(x, y) \, dx \, dy &= \iint_S f(r \cos \theta, r \sin \theta) \left| \frac{\partial(x, y)}{\partial(r, \theta)} \right| \, dr \, d\theta \\ &= \int_{\alpha}^{\beta} \int_a^b f(r \cos \theta, r \sin \theta) \, r \, dr \, d\theta \end{aligned}$$

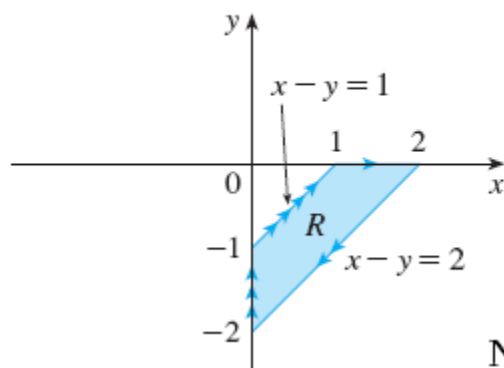
An Example

EXAMPLE 3 Evaluate the integral $\iint_R e^{(x+y)/(x-y)} dA$, where R is the trapezoidal region with vertices $(1, 0)$, $(2, 0)$, $(0, -2)$, and $(0, -1)$.

$$\frac{3}{4}(e - e^{-1})$$



T ↓
↑ T^{-1}

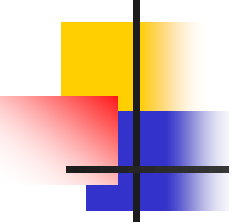


NOTE: In case it is more convenient to express u and v in terms of x and y , we can first compute $\frac{\partial(u, v)}{\partial(x, y)}$ explicitly and then find the needed Jacobian $\frac{\partial(x, y)}{\partial(u, v)}$ from the formula

$$\frac{\partial(x, y)}{\partial(u, v)} \cdot \frac{\partial(u, v)}{\partial(x, y)} = 1,$$

which is a consequence of the Chain Rule.

Triple Integrals



There is a similar change of variables formula for triple integrals. Let T be a transformation that maps a region S in uvw -space onto a region R in xyz -space by means of the equations

$$x = g(u, v, w) \quad y = h(u, v, w) \quad z = k(u, v, w)$$

The **Jacobian** of T is the following 3×3 determinant:

$$\boxed{12} \quad \frac{\partial(x, y, z)}{\partial(u, v, w)} = \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} & \frac{\partial x}{\partial w} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} & \frac{\partial y}{\partial w} \\ \frac{\partial z}{\partial u} & \frac{\partial z}{\partial v} & \frac{\partial z}{\partial w} \end{vmatrix}$$

Under hypotheses similar to those in Theorem 9, we have the following formula for triple integrals:

$$\boxed{13} \quad \iiint_R f(x, y, z) \, dV = \iiint_S f(x(u, v, w), y(u, v, w), z(u, v, w)) \left| \frac{\partial(x, y, z)}{\partial(u, v, w)} \right| du \, dv \, dw$$

V EXAMPLE 4 Use Formula 13 to derive the formula for triple integration in spherical coordinates.